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OPTIMAL BACKWARD ERROR OF A TOTAL LEAST SQUARES AND ITS RANDOMIZED ALGORITHMS*

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Abstract. In this paper, we propose a new randomized iterative algorithm RQI-SPCGTLS (Rayleigh quotient iteration with sketching preconditioned conjugate gradient method for total least squares problems) for solving the large-scale overdetermined total least squares (TLS) problems. In order to reduce the cost of initial guess construction, we prove the effectiveness of a backward stable least squares (LS) solution and utilize the randomized solver for the LS problem. We derive a new explicit expression for the optimal backward error of a TLS system and relate it to the well-known result in the least squares setting. This work formally provides theoretical analysis on the feasibility of leveraging the LS information to solve the TLS problem. As for the preconditioned conjugate gradient (PCG) subroutine, we innovate by substituting the complete Cholesky factorization with the sketching preconditioner. We verify its effectiveness within the finite-precision arithmetic with respect to the reduced condition number and the preservation of the convergence rate. Numerical experiments show that the RQI-SPCGTLS beats the classic RQI-PCGTLS and its mixed precision variant and is likely to be a stable solver when effective.

Key words. randomized iterative methods, total least squares, perturbation analysis, optimal backward error, sketching preconditioner

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1. Introduction.

In recent years, randomized numerical linear algebra has attracted much attention due to its efficiency in solving large-scale scientific computing problems [27, 30, 34]. Successful applications include low-rank matrix approximation [9], linear system solving [23], eigenvalue problems [37], etc.

Total least squares (TLS) problems, also known as errors-in-variables (EIV) models, are widely used in many fields [48, 49]. They handle the errors in all variables and thus are more realistic than their least squares (LS) counterparts. However, while randomized algorithms for LS problems have been extensively studied [1, 17, 58], research into TLS problems remains relatively underdeveloped.

In this paper, we will study a randomized algorithm based on a Rayleigh quotient iteration (RQI) framework to solve the overdetermined TLS problem with a single right-hand side, formulated as

$$\min_{\Delta A,\Delta b}\|[\Delta A,\Delta b]\|_{\mathrm{F}},\ \text{s.t.}\ (A+\Delta A)x=b+\Delta b,\ \text{where}\ A\in\mathbb{R}^{m\times n}, m\gg n. \tag{1.1}$$

The exact solution to the TLS problem (1.1) can be obtained by the singular value decomposition (SVD) of the augmented matrix $B=[A,b]\in\mathbb{R}^{m\times(n+1)}$, where A is

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Algorithm 1.1. A general framework for the TLS problem.

Require: Matrix $A \in \mathbb{R}^{m\times n}$, vector $b \in \mathbb{R}^m$, subroutine INITIALTOOL, subroutine PCGSOLVER
Ensure: TLS solution $x_{\mathrm{TLS}} \in \mathbb{R}^n$

$x_0 \leftarrow$ INITIALTOOL(A, b) ▷ Construct the initial guess
$r_0 = b - Ax_0$
$\sigma_0^2 = r_0^\top r_0/(x_0^\top x_0 + 1)$ ▷ Initial eigenvalue guess of $B \equiv [A, b]$
$u_0 \leftarrow$ PCGSOLVER($A^\top A u_0 = x_0$) ▷ One step of the inverse iteration
$x_1 = x_0 + \sigma_0^2 u_0$ ▷ Ready for RQI
for $k = 1, 2, \ldots$ **do**
  $r_k = b - Ax_k$
  $\sigma_k^2 = r_k^\top r_k/(x_k^\top x_k + 1)$
  $f_k = -A^\top r_k - \sigma_k^2 x_k$
  $g_k = -b^\top r_k + \sigma_k^2$
  $\omega_k \leftarrow$ PCGSOLVER($(A^\top A - \sigma_k^2 I)\omega_k = -f_k$)
  $z_k = x_k + \omega_k$
  $\beta_k = (z_k^\top f_k - g_k)/(z_k^\top x_k + 1)$
  $u_k \leftarrow$ PCGSOLVER($(A^\top A - \sigma_k^2 I)u_k = x_k$)
  $x_{k+1} = z_k + \beta_k u_k$
end for

the data matrix and b is the right-hand side vector [21]. Here and throughout, we denote the singular values of B as $\sigma_1' \geq \sigma_2' \geq \cdots \geq \sigma_{n+1}'$, with the corresponding right singular vectors as $v_i', i = 1, 2, \ldots, n+1$, and denote the singular values of A as $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$, with the corresponding right singular vectors as $v_i, i = 1, 2, \ldots, n$. Throughout the paper, we assume that $\sigma_n > \sigma_{n+1}'$, which guarantees the existence and uniqueness of the TLS solution [21]. By the Eckart–Young–Mirsky theorem [50], the TLS solution is given by

$$\min_{\operatorname{rank}(A+\Delta A, b+\Delta b)<n+1} \|[\Delta A, \Delta b]\|_{\mathrm{F}} = \sigma_{n+1}', \text{ with } x_{\mathrm{TLS}} = -\frac{v_{n+1}'[1:n]^\top}{v_{n+1}'[n+1]}. \tag{1.2}$$

For large-scale problems, the SVD can be computationally prohibitive. A practical alternative is to employ iterative methods. An algorithm due to [3], called the RQI-PCGTLS (Rayleigh quotient Iiteration preconditioned conjugate gradient method for total least squares problems), is effective. It combines the Rayleigh quotient iteration (RQI) method with the preconditioned conjugate gradient (PCG) method. We abstract it in Algorithm 1.1.

For the classic RQI-PCGTLS, the INITIALTOOL is a direct method for LS problems, e.g., the QR decomposition, and the PCGSOLVER uses the R factor from the Cholesky factorization of $A^\top A$ as a preconditioner.

At its core, Algorithm 1.1 solves the eigenvalue problem

$$\begin{bmatrix} A^\top A & A^\top b \\ b^\top A & b^\top b \end{bmatrix} \begin{bmatrix} x \\ -1 \end{bmatrix} = \lambda \begin{bmatrix} x \\ -1 \end{bmatrix} \tag{1.3}$$

to find $x = x_{\mathrm{TLS}}$ and $\lambda = \sigma_{n+1}'^2$. The Rayleigh quotient under this system for each x is computed as $\rho(x) = \|b - Ax\|^2/(1 + \|x\|^2)$. Meanwhile, the normal equation for the TLS problem (1.1) can be obtained from the first row of (1.3) by taking $\lambda = \sigma_{n+1}'^2$,

$$(A^\top A - \sigma_{n+1}'^2 I)x = A^\top b. \tag{1.4}$$

It is the subtraction of $\sigma_{n+1}'^2 I$ that makes TLS problems more challenging than LS problems from many aspects, such as the normal condition number [29, 32]. We also refer the reader to [14, 15, 36, 56, 57] for additional relevant work on other condition numbers and error analysis for TLS problems. To ensure the convergence of the RQI, one step of the inverse iteration is used right after the initial guess is constructed. Here, we omit the detailed iteration procedure, which has been simplified in [3] and used in Algorithm 1.1.

1.1. Motivation and overview of main results. In this paper, we aim to address two computational bottlenecks in Algorithm 1.1: proposing a lower-cost yet effective INITIALTOOL, and providing an efficient preconditioner for the PCG-SOLVER. Our original ideas lie in *utilizing the LS information in a provably reasonable way to solve the TLS more easily.* While our theory and algorithm design are knotted together, we recommend focusing on the theoretical results, whose potential value is not restricted to the algorithm itself. They extend and complement much well-known research [6, 50] from the past decades.

For the INITIALTOOL subroutine, computing the LS solution to construct the initial guess may incur a high and nonnegligible computational cost. Traditional methods, such as the QR decomposition, have a complexity of $\mathcal{O}(mn^2)$. Some existing algorithms overlook this issue, causing this step to be the cost-dominant one.

In section 2, we present a theoretical analysis, demonstrating the advantages of using the backward stable LS solution to solve the TLS problem in certain cases and revealing deeper connections between the LS and TLS systems.

Componentwise error bound. The componentwise error of a backward stable LS solution $\hat{x}$ to the true TLS solution x_{TLS} with respect to the right singular vectors of A is well upper-bounded for $i=1,2,\ldots,n$ as

$$\left| v_i^\top \left(\left(1 - \frac{\sigma_{n+1}'^2}{\sigma_i^2}\right) x_{\mathrm{TLS}} - \hat{x} \right) \right| \lesssim \frac{1}{\sigma_i} \|\hat{x}\| \|[A,b]\| u + \frac{1}{\sigma_i^2} \|b - A\hat{x}\| \|[A,b]\| u,$$

where u is the unit roundoff and $\|\cdot\|$ denotes the 2-norm of a vector or the spectral norm of a matrix. The notation $\lesssim$ means that the left-hand side is bounded above by $C(m,n)$ times the right-hand side for some low-degree polynomial $C(m,n)$. This result is especially beneficial when σ_{n+1}'/σ_i is away from 1 (see Theorem 2.2). In Propositions 2.4 and 2.5, we point out that the residual of the true LS solution is crucial to this ratio, and an effective condition number is a better indicator of the difficulty of a TLS problem, if starting from an LS solution.

Bound for the distance to the nearest compatible system. The distance δ from a computed backward stable LS solution $\hat{x}$ to its nearest compatible system, i.e.,

$$\delta = \min \|[E,f]\|_{\mathrm{F}}, \text{ s.t. } (A+E)\hat{x} = b + f,$$

is already moderate, upper-bounded as $\delta \lesssim \kappa u \frac{\|b\| + \|A\|\|\hat{x}\|}{(1+\|\hat{x}\|^2)^{1/2}}$, where $\kappa u \ll 1$ and $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm. This result supports the choice of utilizing backward stable LS solutions in TLS problems (see Theorem 2.7).

Optimal backward error for TLS. A newly derived expression for the optimal backward error of a computed TLS solution $\tilde{x}$ is given by

$$\min_{E,e} \|[E,e]\|_{\mathrm{F}}^2 = \frac{\|\tilde{r}\|^2}{1+\|\tilde{x}\|^2} + \min\left\{0, \lambda_{\min}\left(\tilde{A}\tilde{A}^\top - \frac{\tilde{r}\tilde{r}^\top}{1+\|\tilde{x}\|^2}\right)\right\}, \tag{1.5}$$

where $\tilde{A} = A + b\tilde{x}^\top$ and $\tilde{r} = b - A\tilde{x}$ (see Theorem 2.9). This formula is mathematically equivalent to the well-known result in [6] but with a different method, whose philosophy can be traced back to the least squares. When considering the errors in both

A and b in an LS problem, we see that a computed LS solution $\hat{x}$ has the optimal backward error [51] as

$$\min_{E,e} \|[E,e]\|_F^2 = \frac{\|\hat{r}\|^2}{1+\|\hat{x}\|^2} + \min\left\{0, \lambda_{\min}\left(A^\top A - \frac{\hat{r}\hat{r}^\top}{1+\|\hat{x}\|^2}\right)\right\}, \tag{1.6}$$

where $\hat{r} = b - A\hat{x}$. Obviously, (1.5) and (1.6) differ only in the matrices $\tilde{A} = A + b\tilde{x}^\top$ and A, where the TLS includes the effects of b and $\tilde{x}$. Thus, the LS and TLS problems are closely related in this way. An application is given in Figure 3b.

These instructive results provide a theoretical explanation for the numerical behavior in which one step of the inverse iteration from a backward stable LS solution may suffice for convergence. Motivated by this insight, we propose using randomized algorithms to efficiently compute the initial guess, provided the backward stability can be ensured. In our design, we employ the recently proposed FOSSIL algorithm [19] which is provably backward stable and of $\mathcal{O}(mn\log(n/u) + n^3)$ complexity.

For the PCGSOLVER subroutine, a widely used preconditioner, the complete Cholesky factorization, is not efficient enough for large-scale problems and sometimes leads to divergence (see Figure 2). Recall that our goal is to

solve the linear system $(A^\top A - \sigma^2 I)x = y$ by PCG fast and accurately,

where σ is zero for the inverse iteration or the current Rayleigh quotient for others, and x, y are subjected to the specific needs of the algorithm. The complete Cholesky factor R of $A^\top A$, or R from a QR decomposition of A, preconditions the system into an equivalent one, i.e.,

$$(I - \sigma^2 R^{-\top} R^{-1})s = R^{-\top} y, \quad x = R^{-1}s. \tag{1.7}$$

The condition number of (1.7) is reduced by a factor of $\kappa(A)$ [3].

In this paper, we employ the sketching technique as an alternative to achieve similar performance. Denoting $S \in \mathbb{R}^{d\times n}$ as a sketching matrix for A (see Definition 3.1) and obtaining R by the QR decomposition of SA, we follow the procedure of

$$SA = QR, \quad (R^{-\top} A^\top A R^{-1} - \sigma^2 R^{-\top} R^{-1})s = R^{-\top} y, \quad x = R^{-1}s. \tag{1.8}$$

Combined with the PCG method, the subroutine is called SPCG (sketching preconditioned conjugate gradient) and shown in Algorithm 1.2.

Note that `pcgsystem` is a function that computes the matrix-vector multiplication

$$(R^{-\top} A^\top A R^{-1} - \sigma^2 R^{-\top} R^{-1})y$$

by the triangular solver. In section 3, we prove the effectiveness of our sketching technique within the finite precision from two aspects.

Sketching preconditioner lowers the condition number. Assuming that

1. $\eta \le 0.9$ *(useful sketching)*, $\|\mathrm{fl}(SA) - SA\| \lesssim \|A\|u$ *(stable sketching)*;
2. $\kappa u \ll 1$ *(A is numerically full-rank)*;
3. σ'_{n+1}/σ_n is away from 1 *(a generous assumption)*;

we have that the condition number of the preconditioned system (1.8) is clustered around 1 (see Theorem 3.3).

The convergence rate is maintained. Following the above assumptions, we consider the inverse iteration starting from a backward stable x_{LS} and the shift $\sigma = 0$. Let $B \equiv [A, b]$ be the augmented matrix and $B^\top B y = x + g$ be the computed situation.

Algorithm 1.2. SPCG: Sketching preconditioned conjugate gradient method.

Require: Matrix $A \in \mathbb{R}^{m\times n}$, vector $y \in \mathbb{R}^n$, parameter $\sigma \in \mathbb{R}$
Ensure: Solution $x \in \mathbb{R}^n$ to $(A^\top A - \sigma I)x = y$
Drawing a sketching matrix S for A
$(Q, R) \leftarrow \mathrm{QR}(SA)$ ▷ can be implemented in single precision in practice
$c(1), \delta y(1) \leftarrow R^{-\top} y$ ▷ use the triangular solver
$p(1), r(1) \leftarrow c(1) - \mathbf{pcgsystem}(\delta y(1), A, R, \sigma)$, $\mathrm{rsq}(1) \leftarrow r(1)^\top r(1)$
for $i = 1, 2, \ldots$ **do**
  $\mathrm{Mp}(i) \leftarrow \mathbf{pcgsystem}(\delta y(i), A, R, \sigma)$
  $\alpha(i) \leftarrow \mathrm{rsq}(i)/(p(i)^\top \mathrm{Mp}(i))$
  $\Delta(i) \leftarrow \alpha(i) \cdot p(i)$
  $\delta y(i+1) \leftarrow \delta y(i) + \Delta(i)$
  $r(i+1) \leftarrow r(i) - \alpha(i) \cdot \mathrm{Mp}(i)$
  $\mathrm{rsq}(i+1) = r(i+1)^\top r(i+1)$
  $\beta(i) \leftarrow \mathrm{rsq}(i+1)/\mathrm{rsq}(i)$
  $p(i+1) \leftarrow r(i+1) + \beta(i) \cdot p(i)$
end for

Algorithm 1.3. RQI-SPCGTLS.

Require: Matrix $A \in \mathbb{R}^{m\times n}$, vector $b \in \mathbb{R}^m$
Ensure: TLS solution $x_{\mathrm{TLS}} \in \mathbb{R}^n$
$x_0 \leftarrow \mathrm{FOSSIL}(A, b)$ ▷ $\mathcal{O}(\mathbf{mn} \cdot \mathbf{log}\,(\mathbf{n/u}) + \mathbf{n^3})$
$r_0 = b - Ax_0$ ▷ $\mathbf{2mn + m}$
$\sigma_0^2 = r_0^\top r_0/(x_0^\top x_0 + 1)$ ▷ $\mathbf{2m + 2n - 2}$
$u_0 \leftarrow \mathrm{SPCG}(A, x_0, 0)$ ▷ $\mathcal{O}(\mathbf{mn} \cdot \mathbf{log}\,(\mathbf{n/u}) + \mathbf{n^3})$
$x_1 = x_0 + \sigma_0^2 u_0$ ▷ $\mathbf{2n}$
for $k = 1, 2, \ldots$ **do**
  $r_k = b - Ax_k$ ▷ $\mathbf{2mn + m}$
  $\sigma_k^2 = r_k^\top r_k/(x_k^\top x_k + 1)$ ▷ $\mathbf{2m + 2n - 2}$
  $f_k = -A^\top r_k - \sigma_k^2 x_k$ ▷ $\mathbf{2mn + 2n}$
  $g_k = -b^\top r_k + \sigma_k^2$ ▷ $\mathbf{2m - 1}$
  $\omega_k \leftarrow \mathrm{SPCG}(A, -f_k, \omega_k)$ ▷ $\mathcal{O}(\mathbf{mn} \cdot \mathbf{log}\,(\mathbf{n/u}) + \mathbf{n^3})$
  $z_k = x_k + \omega_k$ ▷ $\mathbf{n}$
  $\beta_k = (z_k^\top f_k - g_k)/(z_k^\top x_k + 1)$ ▷ $\mathbf{4n - 2}$
  $u_k \leftarrow \mathrm{SPCG}(A, x_k, \sigma_k)$
  $x_{k+1} = z_k + \beta_k u_k$ ▷ $\mathbf{2n}$
end for

Denote ϕ as the true angle between g and the target v'_{n+1} and η as the error angle between $(B^\top B)^{-1} g$ and v'_{n+1}. Then we have

$$\tan\eta \leq \frac{\sigma'^2_{n+1}}{\sigma'^2_n} \tan\phi,$$

where $\tan\phi$ is not big due to the backward stability of x_{LS} as discussed. Therefore, the computing error does not change the converging direction much.

Overall, we summarize the above ideas into a new algorithm RQI-SPCGTLS (Rayleigh quotient iteration with sketching preconditioned conjugate gradient method for total least squares problems), shown in Algorithm 1.3. Since the dominant costs happen in the INITIALTOOL and PCGSOLVER subroutines, which are both restricted within $\mathcal{O}(mn\log(n/u)+n^3)$, we conclude that the RQI-SPCGTLS is also of $\mathcal{O}(mn\log(n/u)+n^3)$ complexity.

1.2. Outline. Section 2 investigates how the backward stability of an LS solution contributes to its approximation to the true TLS solution. The relationship between the LS and TLS problems is explored through the lens of the effective condition number and the optimal backward error. Section 3 introduces the sketching preconditioner, demonstrating the effectiveness in finite precision and proving the preservation of the convergence rate. Section 4 provides detailed information on the practical implementation of our proposed algorithm. Section 5 presents numerical experiments that validate the efficiency of our method. These experiments include complexity verification and comparisons with the state-of-the-art RQI-PCGTLS algorithm and its mixed-precision variant, the RQI-PCGTLS-MP (see Figure 2). Finally, section 6 summarizes our findings and discusses potential avenues for further research.

1.3. Notation. $\|\cdot\|$ denotes the Euclidean norm of a vector or the spectral norm of a matrix, while $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm. The Moore–Penrose pseudoinverse [53] of a matrix A is written as $A^\dagger$, while that of any vector $v\in\mathbb{R}^n$ is denoted by

$$v^\dagger \equiv \begin{cases} 0 & \text{if } v=0,\\ v^\top/\|v\|^2 & \text{if } v\neq 0.\end{cases}$$

The condition number of a matrix A is defined as $\kappa(A)\equiv\sigma_{\max}(A)/\sigma_{\min}(A)=\|A\|\|A^\dagger\|$. Without specific declaration, $\kappa(A)$ is simplified as κ. The symbol $a\lesssim b$ means that there exists a polynomial of a low degree $C(m,n)$ such that $a\le C(m,n)b$. u denotes the unit roundoff ($u\approx 10^{-16}$ in double precision). Throughout the paper, we assume that $\kappa u\ll 1$. The numerically computed quantity is denoted by $\mathrm{fl}(\cdot)$, and the error is $\mathrm{err}(f)\triangleq\mathrm{fl}(f)-f$.

2. Initial guess from backward stable least squares solutions. It is suggested from empirical experiments that we can use the aforementioned least squares (LS) solutions to construct the initial guess for the Rayleigh quotient iteration (RQI) when solving the total least squares (TLS) problem. On one hand, to the best of our knowledge, there has been little theoretical insight into why LS solutions can help. On the other hand, large-scale problems urge us to efficiently obtain the initial guess without losing too much accuracy. Thus, a natural question could be the following:

Can we identify specific conditions under which the LS solution becomes advantageous for solving the problem and can be computed more efficiently?

This section provides an affirmative answer to the above question, and for the first time systematically establishes the theoretical foundation necessary for transferring results from the LS to the TLS problems.

2.1. Backward stability makes the LS solution close to what we want. TLS problems aim to minimize perturbations, while backward stability inherently limits the distance within which the computed LS solution can be fitted into a nearby system. The similarity between these two formulations motivates a deeper analysis of their connections.

First, we formally define the backward stability [28] of a least squares problem $Ax\approx b$ when considering the perturbations in both A and b.

DEFINITION 2.1. *A numerically computed least squares solution $\hat{x}$ is backward stable if it is the exact solution to a slightly perturbed problem, i.e.,*

$$\hat{x} = \arg\min_x \|(b+\Delta b) - (A+\Delta A)x\|, \text{ where } \|\Delta A\| \lesssim \|A\|u, \|\Delta b\| \lesssim \|b\|u.$$

A solver is backward stable if it always computes a backward stable solution.

For least squares problems, denote the weighted backward error for the computed $\hat{x}$ as

$$\mu(\hat{x}) = \min_{E,f} \left\{ \|E, \theta f\|_{\mathrm{F}} : (A+E)^\top (A+E)\hat{x} = (A+E)^\top (b+f) \right\}.$$

For the case where $\theta = 1$, a well-known estimate first derived by Karlson and Waldén [31] is given as

$$\nu(\hat{x}) = \left\| \left(\|\hat{x}\|^2 A^\top A + \|b - A\hat{x}\|^2 I \right)^{-\frac{1}{2}} A^\top (b - A\hat{x}) \right\|. \tag{2.1}$$

The accuracy of this estimate has been much discussed. Denote x_{LS} as the true solution to the LS problem which satisfies the normal equation $A^\top A x_{\mathrm{LS}} = A^\top b$. Karlson and Waldén [31] gives the lower bound of $\mu/\nu \geq 2 - \sqrt{2}$. Gu [26] further improves the result to $2/(1-\sqrt{5}) \leq \mu/\nu \leq \|b - A\hat{x}\|_2 / \|b - Ax_{\mathrm{LS}}\|_2$. In [25], Grcar shows that ν is asymptotically optimal in the sense that $\lim_{\hat{x}\to x_{\mathrm{LS}}} \mu/\nu = 1$. Gratton, Jiránek, and Titley-Peloquin [24] derive new bounds that give a satisfactory explanation for the behavior of the estimate. It is also indicated that the Frobenius norm of the error is a small constant multiple of its spectral norm, and thus we can use either to simplify the analysis.

Therefore, $\hat{x}$ is backward stable if and only if $\nu(\hat{x}) \lesssim \|[A, b]\|u$, and we can safely use the computable formula (2.1) to perform our derivation.

Now, we first give the componentwise error between the backward stable LS solution and the true TLS solution with respect to the singular vectors of A.

THEOREM 2.2 (componentwise error). *Let x_{TLS} and $\hat{x}$, respectively, be the true TLS solution and the backward stable solution to the aforementioned LS problem. Then, for $i = 1, 2, \ldots, n$, we have*

$$\left| v_i^\top \left(\left(1 - \frac{\sigma_{n+1}'^2}{\sigma_i^2}\right) x_{\mathrm{TLS}} - \hat{x} \right) \right| \lesssim \frac{1}{\sigma_i} \|\hat{x}\| \|[A, b]\| u + \frac{1}{\sigma_i^2} \|b - A\hat{x}\| \|[A, b]\| u,$$

where v_i is the ith right singular vector of A.

Proof. Denote the singular values of $B \equiv [A, b]$ as $\sigma_1' \geq \sigma_2' \geq \cdots \geq \sigma_{n+1}'$ and those of A as $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$, with the corresponding right singular vectors v_i. From the normal equation of the TLS problem (1.4), we have

$$(A^\top A - \sigma_{n+1}'^2 I) x_{\mathrm{TLS}} = A^\top b \Longleftrightarrow A^\top b + \sigma_{n+1}'^2 x_{\mathrm{TLS}} = A^\top A x_{\mathrm{TLS}}. \tag{2.2}$$

Write x_{TLS} in terms of the right singular vectors of A as $x_{\mathrm{TLS}} = \sum_{i=1}^n (v_i^\top x_{\mathrm{TLS}}) v_i$. Plugging in (2.2) and multiplying $v_i^\top$ on both sides, we get, for $i = 1, 2, \ldots, n$,

$$v_i^\top (A^\top b + \sigma_{n+1}'^2 x_{\mathrm{TLS}}) = v_i^\top A^\top A x_{\mathrm{TLS}} \Rightarrow \sigma_i u_i^\top b + \sigma_{n+1}'^2 v_i^\top x_{\mathrm{TLS}} = \sigma_i^2 v_i^\top A x_{\mathrm{TLS}}.$$

Here, we use the identity that states $Av_i = \sigma_i u_i$, where u_i is the ith left singular vector of A. Thus, we have

$$(\sigma_i^2 - \sigma_{n+1}'^2) v_i^\top x_{\mathrm{TLS}} = \sigma_i u_i^\top b, \quad i = 1, 2, \ldots, n. \tag{2.3}$$

Using (2.3), we can derive that

$$A^\top (b - A\hat{x}) = \sum_{i=1}^{n} \sigma_i \left(u_i^\top b - \sigma_i (v_i^\top \hat{x})\right) v_i$$
$$= \left(\sum_{i=1}^{n} \sigma_i^2 v_i^\top (x_{\rm TLS} - \hat{x}) v_i\right) - \sigma_{n+1}'^2 \left(\sum_{i=1}^{n} v_i^\top x_{\rm TLS} v_i\right).$$

Furthermore, some algebraic manipulations yield

$$\left(\|\hat{x}\|^2 A^\top A + \|b - A\hat{x}\|^2 I\right)^{-\frac{1}{2}} = \sum_{i=1}^{n} \left(\|\hat{x}\|^2 \sigma_i^2 + \|b - A\hat{x}\|^2\right)^{-\frac{1}{2}} v_i v_i^\top.$$

Plugging the above results in the formula (2.1) of $\nu(\hat{x})$, we have

$$\nu(\hat{x})^2 = \sum_{i=1}^{n} \frac{\sigma_i^4}{\|\hat{x}\|^2 \sigma_i^2 + \|b - A\hat{x}\|^2} \left[v_i^\top \left(1 - \frac{\sigma_{n+1}'^2}{\sigma_i^2}\right) x_{\rm TLS} - \hat{x} \right]^2.$$

Thus, $\nu(\hat{x}) \lesssim \|[A, b]\| u$ is equivalent to

$$\left| v_i^\top \left(\left(1 - \frac{\sigma_{n+1}'^2}{\sigma_i^2}\right) x_{\rm TLS} - \hat{x} \right) \right| \lesssim \frac{1}{\sigma_i} \|\hat{x}\| \|[A, b]\| u + \frac{1}{\sigma_i^2} \|b - A\hat{x}\| \|[A, b]\| u \forall i.$$

Then we complete the proof. ∎

Theorem 2.2 is attractive when σ_{n+1}'/σ_n is away from 1. Note that until now, the existing RQI-PCGTLS algorithm [3], its mixed-precision variant RQI-PCGTLS-MP [38], and some randomized algorithms [55] all assumed this condition to guarantee a good performance (see Figure 2). We also refer the reader to other randomized algorithms, such as that proposed in [13], which do not emphasize this condition but leverage the sparse structure, addressing a range of practical considerations complementary to our approach. We reveal the essence of σ_{n+1}'/σ_n later in Proposition 2.4.

We remark that for the true LS solution $x_{\rm LS}$, similar results can be written as

$$\left| v_i^\top (x_{\rm LS} - \hat{x}) \right| \lesssim \frac{1}{\sigma_i} \|\hat{x}\| \|[A, b]\| u + \frac{1}{\sigma_i^2} \|b - A\hat{x}\| \|[A, b]\| u, \quad i = 1, 2, \ldots, n,$$

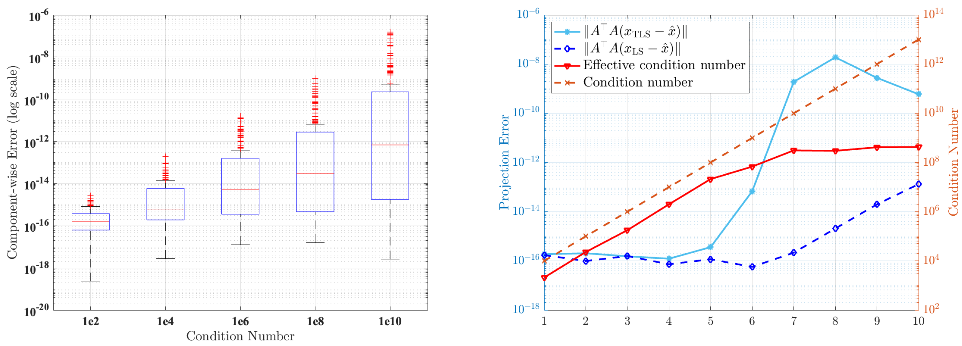
which was already discussed in the literature [19]. It is then natural to derive the projection error bounds.

Corollary 2.3 (projection error). *Let $x_{\rm LS}$ (the true LS solution), $x_{\rm TLS}$ (the true TLS solution), and $\hat{x}$ (the computed backward stable LS solution) be as above. Then we have*

$$\begin{cases} \|A^\top A(x_{\rm LS} - \hat{x})\| \lesssim \|A\|(\|b\| + \|A\|\|\hat{x}\|) u, \\ \|A^\top A(x_{\rm TLS} - \hat{x})\| \lesssim \sigma_{n+1}'^2 \|x\| + \|A\|(\|b\| + \|A\|\|\hat{x}\|) u. \end{cases} \tag{2.4}$$

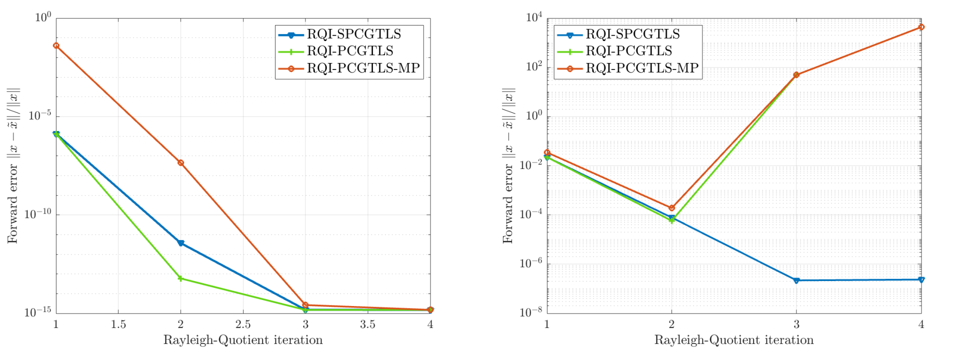
Notice that the RQI method for the TLS problem is essentially tackling an eigenvalue problem for $B^\top B$, where $B \equiv [A, b]$. In fact, denote $y = [x_{\rm TLS}^\top, -1]^\top, \hat{y} = [\hat{x}^\top, -1]^\top$, and consider

$$B^\top B(y - \hat{y}) = \begin{pmatrix} A^\top \\ b^\top \end{pmatrix} (A, b) \begin{pmatrix} x_{\rm TLS} - \hat{x} \\ 0 \end{pmatrix} = \begin{pmatrix} A^\top A(x_{\rm TLS} - \hat{x}) \\ b^\top A(x_{\rm TLS} - \hat{x}) \end{pmatrix};$$



(a) Component-wise error (b) Projection error

Fig. 1. *Consider two kinds of errors between the computed backward stable LS solution and the true TLS solution in* 5000×200 *random examples with changing condition numbers.* (a) *is the boxplot of the componentwise error, namely,* $|v_i^\top((1-\sigma_{n+1}'^2/\sigma_i^2)x_{\mathrm{TLS}}-\hat{x})|$. (b) *is the norm of projection error from* (2.4) *with reference of the effective and normal condition numbers; see Remark* 2.6.



(a) 100×96 well-conditioned Toeplitz system (b) 10000×500 ill-conditioned random system

Fig. 2. *When the system tends to be overdetermined and ill-conditioned, the original or mixed-precision RQI-PCGTLS(-MP) fail to give good performance, while our RQI-SPCGTLS still works well.*

then from Corollary 2.3, we know that a backward stable LS solution makes the projection of the initial $\hat{y}\in\mathbb{R}^{n+1}$ approximately close to the projection of the true solution y on n components.

For illustration, we provide random examples of size 5000×200 with changing condition numbers $\kappa(A)\in[10^1,10^{13}]$. Given a fixed condition number, we generate A with logarithmically spaced singular values. Then we create a uniformly random unit vector x and a single right-hand side b with the residual norm $\|b-Ax\|=10^{-8}$. The results are shown in Figure 1. Theorem 2.2 is well demonstrated in Figure 1a, where the componentwise error is small even for ill-conditioned cases.

Figure 1b is insightful. We provide two propositions, previously not formally established, to explain two phenomena observed therein. The first one is the sharp increase occurring at the condition number of 10^8, and the second is the slight oscillation beyond the condition number of 10^{10}.

PROPOSITION 2.4 (the residual of the LS solution matters). *Let $x_{\rm LS}$ be the LS solution and $r = b - Ax_{\rm LS}$ be its residual. Denote $\kappa = \sigma_{\max}(A)/\sigma_{\min}(A)$ as the normal condition number of A and follow the previous notation. If there is a gap between* 1 *and $\kappa\|r\| < 1$, then the ratio σ'_{n+1}/σ_n is away from* 1.

Proof. If $\|A\| > 1$, do the column scaling and let $\tilde{A} = \frac{A}{\|A\|}, \tilde{b} = \frac{\tilde{b}}{\|A\|}$. Then we have $\tilde{r} = b - \tilde{A}x_{\rm LS}$ with $\|\tilde{r}\|\kappa(\tilde{A}) < 1$ and $\|\tilde{A}\| = 1$. Thus, without loss of generality, we can assume $\|A\| = 1$. For $B \equiv [A, b] = [A, Ax_{\rm LS} + r]$, the singular values of B are identical to those of $[A, r]$. By the least squares property, $r \perp {\rm Range}(A)$, namely, $A^\top r = 0$. Thus,

$$B^\top B \sim \begin{pmatrix} A^\top A & A^\top r \\ r^\top A & r^\top r \end{pmatrix} = \begin{pmatrix} A^\top A & 0 \\ 0 & r^\top r \end{pmatrix}.$$

The symbol '$\sim$' denotes the equivalence relation under congruence transformation. Therefore, in the exact arithmetic, $\sigma'_{n+1} = \min\{\sigma_n, \|r\|\}$. By the assumption that $\|r\| < 1/\kappa = \sigma_n$, we know σ'_{n+1}/σ_n is away from 1. ∎

Proposition 2.4 explains σ'_{n+1}/σ_n from the perspective of the LS residual, indicating that the LS solution is informative for the TLS problem. The very closeness of the two blue lines in Figure 1b before the condition number of 10^8 is in alignment with our result. On the other hand, the mentioned sharp increase illustrates the unsatisfactory case where $\kappa\|r\| > 1$. Namely, when the residual is large in the sense of $\kappa\|r\| > 1$, the proximity of the LS solution is diminished.

We show in the next proposition that the effective condition number is a better measure for the TLS problem than the normal condition number if we start from an LS solution. The effective condition number has existed in the literature [4, 8, 33] but has not received enough attention.

PROPOSITION 2.5 (effective condition number). *Assume we are given an LS solution $x_{\rm LS}$ and the true TLS solution $x_{\rm TLS}$, and denote $\Delta x = x_{\rm TLS} - x_{\rm LS}$. Write the effective condition number $\kappa_{\rm eff} = \frac{\|b\|}{\sigma_n \|x_{\rm LS}\|}$ and $\delta = \frac{\sigma'_{n+1}}{\sigma_n}$. Then we have*

$$\frac{\|\Delta x\|}{\|x_{\rm LS}\|} \le \kappa_{\rm eff} \frac{\delta}{1-\delta} \left(\sqrt{2} + \frac{\sigma_n}{\|b\|} \right).$$

Proof. By the formulation of the TLS problem (1.1) and its solution (1.2), we have the perturbation system

$$(A + \Delta A)(x_{\rm LS} + \Delta x) = b + \Delta b, \text{ where } \|[\Delta A, \Delta b]\|_{\rm F} = \sigma'_{n+1}.$$

Evidently, $\|\Delta b\| \le \sigma'_{n+1}$. Taking $\mu = \sqrt{2}$ in Theorem 1.4.3 of [33], in our case we have

$$\frac{\|\Delta x\|}{\|x_{\rm LS}\|} \le \kappa_{\rm eff} \frac{1}{1-\delta} \left(\sqrt{2}\delta + \frac{\|\Delta b\|}{\|b\|} \right),$$

which can be further simplified to

$$\frac{\|\Delta x\|}{\|x_{\rm LS}\|} \le \kappa_{\rm eff} \frac{\delta}{1-\delta} \left(\sqrt{2} + \frac{1}{\delta} \frac{\sigma'_{n+1}}{\|b\|} \right) = \kappa_{\rm eff} \frac{\delta}{1-\delta} \left(\sqrt{2} + \frac{\sigma_n}{\|b\|} \right). \qquad ∎$$

Proposition 2.5 tells us that for a given system (1.1) and a computed backward stable LS solution $x_{\rm LS}$, the effective condition number $\kappa_{\rm eff}$ and the ratio δ play the key roles in the approximation of the true TLS solution.

Remark 2.6. Denote u_i $(i=1,2,\ldots,m)$ as the ith left singular vector of A, and decompose $b=\sum_{i=1}^m \beta_i u_i$. Since $x_{\mathrm{LS}}=A^\dagger b$, we have

$$\|x_{\mathrm{LS}}\|=\sqrt{\sum_{i=1}^n \frac{\beta_i^2}{\sigma_i^2}} \Longrightarrow \kappa_{\mathrm{eff}}=\frac{\|b\|}{\sigma_n\|x_{\mathrm{LS}}\|}=\frac{\sqrt{\beta_1^2+\beta_2^2+\cdots+\beta_n^2}}{\sigma_n\sqrt{\left(\frac{\beta_1}{\sigma_1}\right)^2+\left(\frac{\beta_2}{\sigma_2}\right)^2+\cdots+\left(\frac{\beta_n}{\sigma_n}\right)^2}}.$$

Thus, $\kappa_{\mathrm{eff}}\le\kappa(A)=\sigma_1/\sigma_n$, and the equality holds if and only if $\|b\|=|\beta_1|$. Therefore, κ_{eff} is sharper for the TLS problem. Figure 1b is reasonable from the perspective that when the effective condition number stops increasing, the projection error follows the trend even though the normal condition number is still growing. Actually, κ_{eff} is especially suitable for the TLS problem since it takes both b and x_{LS} into account. We hope our work can provide new insight into this.

For the last part of this section, we show that the perturbation to the nearest compatible system of a backward stable LS solution is well upper-bounded.

THEOREM 2.7 (optimal perturbation norm). *Given a backward stable LS solution $\hat{x}$, consider its nearest compatible system, i.e.,*

$$\delta=\min\|[E,f]\|_{\mathrm{F}},\ \textit{s.t.}\ (A+E)\hat{x}=b+f;$$

then δ is upper-bounded as

$$\delta\lesssim\kappa u\frac{\|b\|+\|A\|\|\hat{x}\|}{(1+\|\hat{x}\|^2)^{1/2}},\ \textit{where}\ \kappa u\ll 1.$$

Proof. By the generalized inverses theory, for any fixed f, define $\hat{r}=b-A\hat{x}$, and then E is in the form of

$$E=(f+\hat{r})\hat{x}^\dagger+Z(I-\hat{x}\hat{x}^\dagger),$$

where Z is an arbitrary matrix. Thus, we turn to minimize

$$\mathcal{L}(f)=\mathrm{tr}(E^\top E)+\|f\|^2$$

over all possible f and Z. By the identities $\hat{x}^\dagger(I-\hat{x}\hat{x}^\dagger)=0$ and $(I-\hat{x}\hat{x}^\dagger)\hat{x}=0$, we can simplify the expression to

$$\mathcal{L}(f)=\frac{\|f+\hat{r}\|^2}{\|\hat{x}\|^2}+\|f\|^2+\|Z(I-\hat{x}\hat{x}^\dagger)\|_{\mathrm{F}}^2\ge\frac{\|f+\hat{r}\|^2}{\|\hat{x}\|^2}+\|f\|^2\triangleq\tilde{\mathcal{L}}(f),$$

with $Z(I-\hat{x}\hat{x}^\dagger)=0$ to achieve the equality. Then, take $\nabla\tilde{\mathcal{L}}(f)=0$, and by some algebra, we obtain that

$$f=-\frac{1}{1+\|\hat{x}\|^2}\hat{r},\quad E=\frac{\|\hat{x}\|^2}{1+\|\hat{x}\|^2}\hat{r}\hat{x}^\dagger.$$

Thus, $\delta=\frac{\|\hat{r}\|}{(1+\|\hat{x}\|^2)^{1/2}}$. Notice that

$$\|A^\top\hat{r}\|\ge\sigma_n\|\hat{r}\|=\|A^\dagger\|^{-1}\|\hat{r}\|,$$

$$\|A^\top\hat{r}\|=\|A^\top(A\hat{x}-b)\|=\|A^\top(Ax_{\mathrm{LS}}-b)+A^\top A(\hat{x}-x_{\mathrm{LS}})\|=\|A^\top A(\hat{x}-x_{\mathrm{LS}})\|,$$

where we use the normal equation of the true LS solution x_{LS}. By Corollary 2.3, we have $\|A^\top A(\hat{x}-x_{\mathrm{LS}})\| \lesssim u\|A\|(\|b\|+\|A\|\|\hat{x}\|)$. Therefore,

$$\|\hat{r}\| \le \|A^\dagger\|\|A^\top \hat{r}\| \lesssim \kappa u\left(\|b\|+\|A\|\|\hat{x}\|\right).$$

Plugging in the expression of δ, we get the desired result. ∎

Theorem 2.7 shows the distance of the computed $\hat{x}$ to the nearest compatible system. By the assumption that $\kappa u \ll 1$, we can expect this quantity to be moderate. Since the TLS problem exactly aims to minimize the perturbation, our choice of $\hat{x}$ as the ingredient for the initial guess construction is beneficial.

2.2. Regarding the optimal backward error for TLS problems. In this section, we propose a new explicit expression for the optimal backward error of the TLS problems. We first briefly review the perturbation analysis for the LS cases here. For the given $A, b, \hat{x}$, and $\hat{r}=b-A\hat{x}$ in a least squares setting, define

$$\mathcal{G}=\left\{[E,e]\in\mathbb{R}^{m\times(n+1)}:\|(b+e)-(A+E)\hat{x}\|_2=\min_{x\in\mathbb{R}^n}\|(b+e)-(A+E)x\|_2\right\};$$

then

$$\min_{[E,e]\in\mathcal{G}}\|[E,e]\|_{\mathrm{F}}^2=\frac{\|\hat{r}\|^2}{1+\|\hat{x}\|^2}+\min\left\{0,\lambda_{\min}\left(AA^\top-\frac{\hat{r}\hat{r}^\top}{1+\|\hat{x}\|^2}\right)\right\}. \tag{2.5}$$

This result, due to Waldén, Karlson, and Sun [51], is obtained by setting the weighting parameter to 1. Our next theorem gives the optimal backward error of a TLS problem exactly in this form but with a little change in the second term. Although the backward error analysis for TLS systems itself is not original, e.g., the famous results in [6], we are unaware of any existing work that highlights the similarity between the LS and the TLS from the perspective we adopt here.

First, it is well known from [21] that if A is of full-rank and b is not orthogonal to the left singular vector subspace of A corresponding to its smallest singular value, the TLS problem (1.1) has the solution x_{TLS} if and only if

$$x_{\mathrm{TLS}} \text{ achieves a global minimum of } \frac{\|b-Ax_{\mathrm{TLS}}\|^2}{1+\|x_{\mathrm{TLS}}\|^2}=\min_y\frac{\|b-Ay\|^2}{1+\|y\|^2};$$

further, if and only if

$$A^\top(b-Ax_{\mathrm{TLS}})=-\frac{\|b-Ax_{\mathrm{TLS}}\|^2}{1+\|x_{\mathrm{TLS}}\|^2}x \text{ and } \frac{\|b-Ax_{\mathrm{TLS}}\|^2}{1+\|x_{\mathrm{TLS}}\|^2}<\sigma_{\min}^2(A), \tag{2.6}$$

here and throughout, we always assume the prerequisites hold. Then, the backward error for a computed solution $\tilde{x}$ can be formally given by

$$\min_{E,e}\|[E,e]\|_{\mathrm{F}}, \text{ s.t. } \frac{\|(b+e)-(A+E)\tilde{x}\|^2}{1+\|\tilde{x}\|^2}=\min_y\frac{\|(b+e)-(A+E)y\|^2}{1+\|y\|^2}. \tag{2.7}$$

By the necessary and sufficient condition (2.6), we know that $[(A+E),(b+e)]$ is bound to be in a feasible set as

$$\mathcal{F}=\left\{[(A+E),(b+e)]\in\mathbb{R}^{m\times(n+1)}: \frac{\|(b+e)-(A+E)\tilde{x}\|^2}{1+\|\tilde{x}\|^2}<\sigma_{\min}^2(A+E)\right.$$
$$\left.\text{and } (A+E)^\top\left((b+e)-(A+E)\tilde{x}\right)=-\frac{\|(b+e)-(A+E)\tilde{x}\|^2}{1+\|\tilde{x}\|^2}\tilde{x}\right\}.$$

An elegant characterization of a possible set $\mathcal{F}^+$ is derived in [5] which we will take advantage of in our analysis.

LEMMA 2.8. *For given* $b \in \mathbb{R}^m$ *and nonzero* $y \in \mathbb{R}^n$, *define*

$$\mathcal{F}^+ = \left\{ F \in \mathbb{R}^{m\times n} : F^\top (b - Fy) = -y \frac{\|b - Fy\|^2}{1 + \|y\|^2} \right\}$$

and

$$\mathcal{N} = \left\{ -\omega\omega^\dagger b y^\top + (I - \omega\omega^\dagger) \left[b y^\dagger + W(I - yy^\dagger) \right] : \omega \in \mathbb{R}^m, W \in \mathbb{R}^{m\times n} \right\};$$

then $\mathcal{F}^+ = \mathcal{N}$.

Now we use this tool to derive an optimal backward error formula for the perturbation problem (2.7).

THEOREM 2.9. *Define* $\tilde{A} = A + b\tilde{x}^\top, \tilde{r} = b - A\tilde{x}$; *then for the backward perturbation problem* (2.7) *of a total least squares system, we have*

$$\min_{E,e} \|[E, e]\|_{\mathrm{F}}^2 = \frac{\|\tilde{r}\|^2}{1 + \|\tilde{x}\|^2} + \min\left\{ 0, \lambda_{\min}\left(\tilde{A}\tilde{A}^\top - \frac{\tilde{r}\tilde{r}^\top}{1 + \|\tilde{x}\|^2} \right) \right\}. \tag{2.8}$$

Proof. For a fixed e, denote $\tilde{b} = b + e$ and $\hat{r} = \tilde{b} - A\tilde{x} = \tilde{r} + e$. Then by Lemma 2.8, we have

$$A + E \in \left\{ -\omega\omega^\dagger \tilde{b}\tilde{x}^\top + (I - \omega\omega^\dagger) \left[\tilde{b}\tilde{x}^\dagger + W(I - \tilde{x}\tilde{x}^\dagger) \right] : \omega \in \mathbb{R}^m, W \in \mathbb{R}^{m\times n} \right\}.$$

By some algebra, we can obtain that

$$(I - \omega\omega^\dagger)\tilde{b}\tilde{x}^\dagger - A = \hat{r}\tilde{x}^\dagger - \omega\omega^\dagger(A + \hat{r}\tilde{x}^\dagger) - (I - \omega\omega^\dagger)A(I - \tilde{x}\tilde{x}^\dagger).$$

The last term can be merged into the arbitrary W, and thus

$$E \in \left\{ \hat{r}\tilde{x}^\dagger - vv^\dagger(A + \tilde{b}\tilde{x}^\top + \hat{r}\tilde{x}^\dagger) + (I - vv^\dagger)Z(I - \tilde{x}\tilde{x}^\dagger) : v \in \mathbb{R}^m, Z \in \mathbb{R}^{m\times n} \right\}.$$

Partition E as $E = K + L$, where

$$K = (I - vv^\dagger)[\hat{r}\tilde{x}^\dagger + Z(I - \tilde{x}\tilde{x}^\dagger)], \quad L = -vv^\dagger(A + \tilde{b}\tilde{x}^\top).$$

From $L^\top K = 0$, we get $\|E\|_{\mathrm{F}}^2 = \|K\|_{\mathrm{F}}^2 + \|L\|_{\mathrm{F}}^2$. Denote $\bar{A} = A + \tilde{b}\tilde{x}^\top$, and by some calculation, we have

$$\|L\|_{\mathrm{F}}^2 = \|vv^\dagger \bar{A}\|_{\mathrm{F}}^2 = \mathrm{tr}(vv^\dagger \bar{A}\bar{A}^\top vv^\dagger) = \mathrm{tr}(v^\dagger \bar{A}\bar{A}^\top v) = v^\dagger \bar{A}\bar{A}^\top v,$$
$$\|K\|_{\mathrm{F}}^2 = \frac{\|\hat{r}\|^2}{\|\tilde{x}\|^2} - \frac{v^\dagger \hat{r}\hat{r}^\top v}{\|\tilde{x}\|^2} + \mathrm{tr}\left[(I - \tilde{x}\tilde{x}^\dagger)Z^\top Z\right] - \mathrm{tr}\left[(I - \tilde{x}\tilde{x}^\dagger)Z^\top vv^\dagger Z\right].$$

Assuming that the eigenvalues for the positive definite matrix $Z(I - \tilde{x}\tilde{x}^\dagger)Z^\top$ are $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, the `trace` part in $\|K\|_{\mathrm{F}}^2$ can be written as

$$\mathrm{tr}\left[Z(I - \tilde{x}\tilde{x}^\dagger)Z^\top\right] - \frac{v^\top}{\|v\|} Z(I - \tilde{x}\tilde{x}^\dagger)Z^\top \frac{v}{\|v\|} \geq \sum_{i=2}^n \lambda_i^2 \geq 0,$$

and the equality (i.e., the minimum) is achieved by $Z = 0$. Therefore, we get the case for a fixed e that

$$\|E\|_{\mathrm{F}}^2 = \frac{\|\hat{r}\|^2}{\|\tilde{x}\|^2} + \min_v \left\{ v^\dagger \left(\bar{A}\bar{A}^\top - \frac{\hat{r}\hat{r}^\top}{\|\tilde{x}\|^2} \right) v \right\}.$$

Thus, we can minimize over all $e \in \mathbb{R}^m$ to get

$$\min_{E,e} \|[E,e]\|_{\mathrm{F}}^2 = \min_e \left\{ \|e\|^2 + \frac{\|\hat{r}\|^2}{\|\tilde{x}\|^2} + \min_y \left\{ y^\dagger \bar{A}\bar{A}^\top y - \frac{y^\dagger \hat{r}\hat{r}^\top y}{\|\tilde{x}\|^2} \right\} \right\}.$$

Notice that

$$\begin{aligned} \bar{A}\bar{A}^\top &= \left[(A + b\tilde{x}^\top) + e\tilde{x}^\top\right] \left[(A + b\tilde{x}^\top) + e\tilde{x}^\top\right]^\top \\ &= (A + b\tilde{x}^\top)(A + b\tilde{x}^\top)^\top + \left[e\tilde{x}^\top (A + b\tilde{x}^\top)^\top + (A + b\tilde{x}^\top)(e\tilde{x}^\top)^\top + \|\tilde{x}\|^2 ee^\top\right] \\ &\triangleq \tilde{A}\tilde{A}^\top + B(e). \end{aligned}$$

Also notice that

$$\|\hat{r}\|^2 - y^\dagger \hat{r}\hat{r}^\top y = \tilde{r}^\top (I - yy^\dagger)\tilde{r} + \tilde{r}^\top (I - yy^\dagger)e + e^\top (I - yy^\dagger)\tilde{r} + e^\top e - e^\top yy^\dagger e.$$

Simplifying the expression by the above two equations, we arrive at

$$\min_{E,e} \|[E,e]\|_{\mathrm{F}}^2 = \min_y \left[y^\dagger \tilde{A}\tilde{A}^\top y + \frac{\tilde{r}^\top (I - yy^\dagger)\tilde{r}}{\|\tilde{x}\|^2} + \min_e h_y(e) \right],$$

where

$$h_y(e) = \frac{1 + \|\tilde{x}\|^2}{\|\tilde{x}\|^2} e^\top e + \frac{\tilde{r}^\top (I - yy^\dagger)e + e^\top (I - yy^\dagger)\tilde{r}}{\|\tilde{x}\|^2} - \frac{e^\top yy^\dagger e}{\|\tilde{x}\|^2} + y^\dagger B(e) y.$$

It is straightforward to verify that for $\bar{e} = -\frac{1}{1+\|\tilde{x}\|^2}(I - yy^\dagger)\tilde{r}$, we have $y^\dagger B(\bar{e}) y = 0$, and at the same time,

$$h_y(e) - h_y(\bar{e}) = \|e - \bar{e}\|^2 + \frac{1}{\|\tilde{x}\|^2} \|(I - yy^\dagger)e - \bar{e}\|^2 \geq 0.$$

Thus, we obtain

$$\min_e h_y(e) = h_y(\bar{e}) = -\frac{1}{1 + \|\tilde{x}\|^2} \frac{\tilde{r}^\top (I - yy^\dagger)\tilde{r}}{\|\tilde{x}\|^2}.$$

Finally, we can conclude that

$$\begin{aligned} &\min_{E,e} \|[E,e]\|_{\mathrm{F}}^2 = \min_y \left\{ y^\dagger \tilde{A}\tilde{A}^\top y + \frac{\tilde{r}^\top (I - yy^\dagger)\tilde{r}}{1 + \|\tilde{x}\|^2} \right\} \\ \Longrightarrow &\min_{E,e} \|[E,e]\|_{\mathrm{F}}^2 = \frac{\|\tilde{r}\|^2}{1 + \|\tilde{x}\|^2} + \min \left\{ 0, \lambda_{\min} \left(\tilde{A}\tilde{A}^\top - \frac{\tilde{r}\tilde{r}^\top}{1 + \|\tilde{x}\|^2} \right) \right\}, \end{aligned}$$

where $\tilde{A} = A + b\tilde{x}^\top$. We complete the proof. $\square$

We point out that the same method can be extended to the scaled total least squares (STLS) problems [39, 40, 57]. Theorem 2.9 provides a more explicit expression for the optimal backward error of a TLS problem than existing results. Setting both the scaling parameter γ and the weighting parameter ω to 1 in [6], the backward error is formulated with the same notation as above as

$$\min_{E,e} \|[E,e]\|_{\mathrm{F}}^2 = \begin{cases} \dfrac{\|\tilde{r}\|^2}{1 + \|\tilde{x}\|^2} & \text{if } \lambda_{\min}(M) \geq 0, \\ \dfrac{\|\tilde{r}\|^2}{1 + \|\tilde{x}\|^2} + \lambda_{\min}(M) = \sigma_{\min}^2(N) & \text{if } \lambda_{\min}(M) < 0, \end{cases} \tag{2.9}$$

where the two matrices N and M are

$$\begin{cases} N = \left[A(I-\tilde{x}\tilde{x}^\dagger), \frac{\|\tilde{r}\|}{\sqrt{1+\|\tilde{x}\|^2}}(I-\tilde{r}\tilde{r}^\dagger), \frac{1}{\sqrt{\|\tilde{x}\|^2+\|\tilde{x}\|^4}}(A\tilde{x}+\|\tilde{x}\|^2 b)\right], \\ M = A(I-\tilde{x}\tilde{x}^\dagger)A^\top - \frac{1}{1+\|\tilde{x}\|^2}\tilde{r}\tilde{r}^\top + \frac{1}{\|\tilde{x}\|^2+\|\tilde{x}\|^4}(A\tilde{x}+\|\tilde{x}\|^2 b)(A\tilde{x}+\|\tilde{x}\|^2 b)^\top, \end{cases}$$

with the error vector being

$$e = \begin{cases} -\dfrac{1}{1+\|\tilde{x}\|^2}\tilde{r} & \text{if } \lambda_{\min}(M) \geq 0, \\ -\dfrac{1}{1+\|\tilde{x}\|^2}\tilde{r} - ww^\top \left[b - \dfrac{2}{1+\|\tilde{x}\|^2}\tilde{r}\right] & \text{if } \lambda_{\min}(M) < 0, \end{cases}$$

where w is the unit eigenvector corresponding to the smallest eigenvalue of M. Actually, our results are in alignment with Chang and Titley-Peloquin [6]—the mathematical expressions are equivalent, and what we offer is a more concise form. We remark that M is not exactly $\tilde{A}\tilde{A}^\top - \tilde{r}\tilde{r}^\top/(1+\|\tilde{x}\|^2)$. Our e can be neatly written as

$$e = -\frac{1}{1+\|\tilde{x}\|^2}(I - yy^\dagger)\tilde{r},$$

where y takes zero or any eigenvector corresponding to the smallest eigenvalue of $\tilde{A}\tilde{A}^\top - \tilde{r}\tilde{r}^\top/(1+\|\tilde{x}\|^2)$ with $\tilde{A} = A + b\tilde{x}^\top$.

Comparing (2.5) with our (2.8) in Theorem 2.9, we can see that the backward error for the TLS problem can be related to that for its corresponding LS problem when considering both perturbations in A and b. In (2.8), $\tilde{A}$ involves both the right-hand side b and the computed $\tilde{x}$, which is not surprising if we think about the nature of the total least squares problem. We expect this new perspective to be useful in future analysis of TLS systems.

Remark 2.10 (a posterior observation). By our RQI-SPCGTLS, if it converges to the true solution, we can conjecture that the computed y is (nearly) backward stable due to the LS backward stability and the analogy between (2.5) and (2.8). Namely, if it is an effective solver, it may well be a good one (see Figure 3b).

3. Sketching preconditioned conjugate gradient subroutine. In the general framework of Algorithm 1.1, two linear systems require solving efficiently. The existing preconditioning methods always use the complete Cholesky factorization of $A^\top A$ or its mixed-precision version to implement this step; both require $\mathcal{O}(mn^2)$.

Randomized numerical linear algebra (RandNLA) [27, 30, 34] provides efficient tools for solving large-scale scientific computing problems, motivating us to develop innovative preconditioning strategies. One of the most significant techniques comes from the sketching matrix which was first formally introduced in [54] to reduce the condition number of a certain system. We refer the reader to abundant research about randomized algorithms for the least squares problems, e.g., [1, 19, 35] and the reference therein, on sketching-and-precondition or iterative sketching philosophy.

The sketching idea itself is not new and is ready to be transferred to the TLS solvers. We present some preliminaries and then prove the effectiveness of the SPCG subroutine in Algorithm 1.2 within the finite arithmetic.

DEFINITION 3.1. *A (random) matrix $S \in \mathbb{R}^{d\times m}$ is called a sketching matrix for a matrix $A^{m\times n}$ with distortion $0 < \eta < 1$ if*

$$(1-\eta)\|Ay\| \leq \|S(Ay)\| \leq (1+\eta)\|Ay\| \textit{ for all } y \in \mathbb{R}^n.$$

For probabilistic sketching matrices, they satisfy the property with high probability.

The following theorem is well known in the literature [54] and is called randomized preconditioning or whitening.

THEOREM 3.2. *Let S be a sketching matrix for a full-rank A with distortion $\eta \in (0,1)$. Consider the Householder QR factorization $SA = QR$, where Q is orthogonal and R is upper triangular. Then,*

$$\frac{1}{1+\eta} \le \sigma_{\min}(AR^{-1}) \le \sigma_{\max}(AR^{-1}) \le \frac{1}{1-\eta}.$$

In particular, the condition number of AR^{-1} is upper-bounded by $(1+\eta)/(1-\eta)$.

In our RQI-SPCGTLS (Algorithm 1.3), after getting the desired R as above, we solve two preconditioned linear systems,

$$R^{-\top} A^\top A R^{-1} x = y, \tag{3.1}$$

$$\left(R^{-\top} A^\top A R^{-1} - \rho R^{-\top} R^{-1}\right) x = y. \tag{3.2}$$

When ρ is close to the true smallest eigenvalue $\sigma_{n+1}'^2$, we will show that under some assumptions, systems (3.1) and (3.2) are both well-conditioned.

THEOREM 3.3 (effectiveness of the sketching preconditioner). *Take $\rho = \sigma_{n+1}'^2$ and assume*

1. $\eta \le 0.9$ *(useful sketching),* $\|fl(SA) - SA\| \lesssim \|A\|u$ *(stable sketching);*
2. $\kappa u \ll 1$ *(A is numerically full-rank);*
3. $\frac{\sigma_{n+1}'}{\sigma_n}$ *is away from* 1 *(a generous assumption).*

Then, the condition numbers of the computed matrices in (3.1) *and* (3.2) *are both clustered around* 1. *Thus, the preconditioned systems are well-conditioned.*

Proof. Let $SA = QR$ be the exact factorization. By the backward stability of the Householder QR factorization [28], there exists E with $\|E\| \lesssim \|SA\|u$ such that $\text{fl}(SA) = SA + E = \tilde{Q}\hat{R}$. Since $\|SA\| = \|R\|$, $\|E\| \lesssim \|R\|u$. Further, we observe that

$$\|R\| = \|(RA^\dagger)A\| = \|(AR^{-1})^\dagger A\| \le \|(AR^{-1})^\dagger\|\|A\| = \frac{\|A\|}{\sigma_{\min}(AR^{-1})} \le (1+\eta)\|A\|.$$

We have $\|E\| \lesssim \|A\|u$. Next, we want to bound $\sigma_{\min}(\hat{R})$. By the Weyl theorem [22] and the identity $\sigma_{\min}(\hat{R}) = \sigma_{\min}(\tilde{Q}\hat{R})$, we have

$$\sigma_{\min}(\hat{R}) \ge \sigma_{\min}(SA) - \|E\| = \sigma_{\min}(R) - \|E\| \ge \frac{(1-\eta)\|A\|}{\kappa} - \|E\| \gtrsim \frac{\|A\|}{20\kappa}.$$

Here, we use the assumption that $\eta \le 0.9, \kappa u \ll 1$ and suppose $u \le 1/20\kappa$, which leads to $\|E\| \lesssim \|A\|/20\kappa$. The second inequality comes from

$$\|R^{-1}\| = \|A^\dagger(AR^{-1})\| \le \frac{\sigma_{\max}(AR^{-1})}{\sigma_{\min}(A)} \le \frac{1}{1-\eta}\frac{1}{\sigma_{\min}(A)}$$
$$\implies \quad \sigma_{\min}(R) = \|R^{-1}\|^{-1} \ge (1-\eta)\frac{\|A\|}{\kappa}.$$

Thus, we can upper-bound $\|A\hat{R}^{-1}\| = \max_{\|x\|=1} \|A\hat{R}^{-1}x\|$ as

$$\begin{aligned}\|A\hat{R}^{-1}\| &\le \frac{1}{1-\eta} \max_{\|x\|=1} \|S(A\hat{R}^{-1})x\| = \frac{1}{1-\eta} \max_{\|x\|=1} \|(\tilde{Q} - E\hat{R}^{-1})x\| \\ &\le \frac{1}{1-\eta}\left(1 + \|E\hat{R}^{-1}\|\right) \le \frac{1}{1-\eta} + 200\kappa u.\end{aligned}$$

The last inequality is due to

$$\frac{1}{1-\eta}\|E\hat{R}^{-1}\| \le \frac{1}{1-\eta}\|E\|\|R^{-1}\| \le \|A\|u\frac{1}{1-\eta}\frac{20\kappa}{\|A\|} \le 200\kappa u.$$

Since $\kappa u \ll 1$, we have $|\sigma_{\max}(A\hat{R}^{-1}) - \frac{1}{1-\eta}| \lesssim \kappa u$. Similarly, we can derive that $|\sigma_{\min}(A\hat{R}^{-1}) - \frac{1}{1+\eta}| \lesssim \kappa u$. Both lead to the beautiful result that the condition number of the preconditioned system (3.1) in finite precision is around 1, namely, well-conditioned.

For the second system (3.2), we can use the above analysis and further write

$$\begin{cases} \lambda_{\min}\left(\hat{R}^{-\top}A^\top A\hat{R}^{-1} - \sigma_{n+1}'^2\hat{R}^{-\top}\hat{R}^{-1}\right) \gtrsim \left(\frac{1}{1+\eta}\right)^2 - \left(\frac{\sigma_{n+1}'}{\sigma_n}\right)^2, \\ \lambda_{\max}\left(\hat{R}^{-\top}A^\top A\hat{R}^{-1} - \sigma_{n+1}'^2\hat{R}^{-\top}\hat{R}^{-1}\right) \le \lambda_{\max}\left(\hat{R}^{-\top}A^\top A\hat{R}^{-1}\right) \lesssim \left(\frac{1}{1-\eta}\right)^2. \end{cases}$$

Provided that σ_{n+1}'/σ_n is away from 1, the condition number of the preconditioned system (3.2) in finite arithmetic is also around 1. ∎

In Theorem 3.3, we derive provably good preconditioners that benefit from not doing the Cholesky factorization. Note that by Proposition 2.4, we implicitly require the system to be fairly compatible for its true LS solution. Fast and accurate solvers for more general cases are still open topics, essentially highlighting the reason why the TLS problems are harder to tackle.

Finally, we establish in the next proposition that by the computed preconditioner $\hat{R}$, the inverse iteration is maintained as it should be.

PROPOSITION 3.4. *Under the assumptions of Theorem* 3.3*, the inverse iteration step in RQI-SPCGTLS (Algorithm* 1.3*) is still of linear convergence or, equivalently, the computing error does not influence the converging direction.*

Proof. The SPCG subroutine for the inverse iteration part is equivalent to solving z from the system

$$\begin{pmatrix} R^{-\top} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} A^\top A & A^\top b \\ b^\top A & b^\top b \end{pmatrix}\begin{pmatrix} R^{-1} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} z \\ -1 \end{pmatrix} = \begin{pmatrix} R^{-\top} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x' \\ -1 \end{pmatrix},$$

where x' is known. Writing $\hat{R}$ as the computed preconditioner, thanks to the proof of errors in multiplication by whitened basis in [19], we know that there exists a vector e with $\|e\| \lesssim u$, such that

$$\mathrm{err}(\hat{R}^{-\top}z) = \|\hat{R}^{-\top}z\|\hat{R}^{-\top}e, \quad \|\mathrm{err}(\hat{R}^{-\top}A^\top A\hat{R}^{-1}z)\| \lesssim \kappa u\|z\|,$$

where $\kappa u \ll 1$ by assumption. Denote $y = [z^\top, -1]^\top, x = [x'^\top, -1]^\top$, and $B \equiv [A, b]$. The above actually tells us that there exist H and f with $\|H\| \ll \|B\|$ and $\|f\| \ll \|x\|$, such that $(B^\top B - H)y = x + f$. Thus,

$$B^\top By = x + f + Hy \triangleq x + g.$$

Denote $e = (B^\top B)^{-1}g$ as the error. We want to analyze the angle between e and the targeted converging vector.

As before, denote v_{n+1}' as the unit singular vector corresponding to σ_{n+1}' of B and $\phi = \angle(g, v_{n+1}')$ as the true angle. Then, by the orthogonal decomposition, we can write g as

$$g = (v_{n+1}'\cos\phi + u\sin\phi)\|g\|, \text{ where } v_{n+1}'^\top u = 0. \tag{3.3}$$

As we discussed in subsection 2.2, the initial vector $x' = x_{\rm LS}$ is not far away from the true one. Thus, we can generously assume that $|\tan\phi| \le 10$. Then,

$$e = (B^\top B)^{-1} g = \left[v'_{n+1}\cos\phi/\sigma'^2_{n+1} + \hat{u}\|(B^\top B)^{-1}u\|\sin\phi\right]\|g\|,$$

where $\hat{u}$ is the normalized version of $(B^\top B)^{-1}u$. From $v'^\top_{n+1}u = 0$, we know that

$$v'^\top_{n+1}(B^\top B)^{-1}u = \left((B^\top B)^{-1}v'_{n+1}\right)^\top u = \frac{1}{\sigma'^2_{n+1}} v'^\top_{n+1}u = 0.$$

Namely, (3.3) is also an orthogonal decomposition. Then, if η denotes the error angle between e and v'_{n+1}, we have

$$\tan\eta = \|(B^\top B)^{-1}u\|\sigma'^2_{n+1}\tan\phi \le \frac{\sigma'^2_{n+1}}{\sigma'^2_n}\tan\phi.$$

Therefore, the error e is almost in the direction of v'_{n+1}. The computational process still converges in the way we desire. ∎

Although the formula of u has y implicitly, we can still confirm the good performance of the inverse iteration: the error e, which could be as large as the exact solution $(B^\top B)^{-1}x$, is nearly in the direction of v'_{n+1}. Our proof follows the philosophy outlined in [41], and the argument for the RQI process adopts a similar approach.

4. Implementation details. This section provides guidance on the implementation details of the RQI-SPCGTLS, such as the choice of random sketching, the termination criterion, and the shift from the inverse iteration to the Rayleigh quotient iteration.

Choice of sketching. We tend to choose the sketching matrix that is efficient for multiplications and has a good distortion property. While there are many choices [34], we prefer to use sparse sign embeddings based on the timing experiments in [16], i.e., a matrix in the form of

$$\mathcal{S} = \xi^{-1/2}[s_1, s_2, \ldots, s_m],$$

whose columns $s_i \in \mathbb{R}^m$ are independent random vectors with ξ nonzero entries placed in uniformly random positions. The nonzero entries are ± 1 with equal probability. A sparse sign embedding is proven to be a sketching matrix for $A \in \mathbb{R}^{m\times n}$ [10] provided

$$d \ge \text{const}\frac{n\log n}{\eta^2}, \quad \xi \ge \text{const}\frac{\log n}{\eta}.$$

Following the discussions in [7, 19, 46], we take $d \approx 12n$ and $\xi = 8$ throughout Algorithm 1.3.

Termination criterion. For the outer RQI loop, we take the termination criteria in [3] where the proof of global convergence is considered. Namely, if we use the variable names in Algorithm 1.1, the normalized residual norm

$$\gamma_k = \left(\frac{\|f_k\|_2^2}{\|x^{(k)}\|_2^2}\right)^{1/2}, \text{ with } \begin{pmatrix} f_k \\ g_k \end{pmatrix} = \begin{pmatrix} -A^\top r^{(k)} - \rho_k x^{(k)} \\ -b^\top r^{(k)} + \rho_k \end{pmatrix},$$

always decreases. Thus, an increase in γ_k should be caused by roundoff, and we can terminate the iteration. For the inner SPCG loop, we point to [45] for understanding that all the iterative methods we use are essentially Newton methods for some (non)linear equations. An appropriate stopping rule should be carefully considered so that the convergence rate of the outer loop is preserved. This problem has been studied since the 1980s [12]. However, in practice, we find that if the iterative method does work, a fixed moderate iteration number of the SPCG subroutine is enough, such as 15 or less. Other choices depending on the outer loop could be found in [3].

Shift from the inverse iteration to the Rayleigh quotient iteration. Szyld [44] sets up a criteria for $p \geq 1$ step(s) of the inverse iteration before switching to the RQI. The explicit condition is not applicable in our case because it needs the knowledge of the spectrum of $B \equiv [A, b]$. Empirical experience shows that one step is often enough [3], but there are still exceptions. For example, among our numerical experiments, the `JGD_BIBD/bibd_20_10` matrix in the SuiteSparse Matrix Collection requires six steps of the inverse iteration to ensure a good performance (see section 5).

We note that from our observation, *more steps of the inverse iteration do not necessarily lead to better performance,* and most of the time, one step is enough. One of the trickiest parts of the iterative methods for the TLS problems is the uncertainty from the eigenvector that the generated sequence converges to. There have been few attempts to directly improve the classic RQI. Friess, Gilbert, and Scheichl[20] perturb the matrix with an imaginary term dependent on the current Rayleigh vector in order to enlarge the distance between the target eigenvalue and others. However, it requires stronger conditions on the initial guess.

5. Numerical illustration. In this section, we demonstrate that our RQI-SPCGTLS outperforms the existing RQI-PCGTLS and RQI-PCGTLS-MP. All the experiments are implemented in MATLAB R2023a on a MacBook Pro with an Apple M2 Pro chip featuring a total of 10 cores and 16 GB of memory. For the mixed-precision algorithm, we use AdvanpixMCT-5.3.7.15933-macOS to perform the half- or single-precision arithmetic.

For our first experiments, we test on dense random matrices with different sizes and condition numbers. We choose a 100×96 Toeplitz system with $\kappa(A) \approx 1$, i.e., well-conditioned. The matrix A is constructed as $A = \bar{T} + E$, where E is a random Toeplitz matrix and $\bar{T} \in \mathbb{R}^{n \times (n-2\omega)}$ is the lower Toeplitz matrix generated by the first row and the first column as

$$t_{i,1} = \begin{cases} \dfrac{1}{\sqrt{2\pi\alpha^2}} \exp\left(-\dfrac{(\omega - i + 1)^2}{2\alpha^2}\right), & i = 1, 2, \ldots, 2\omega + 1 \\ 0 & \text{otherwise;} \end{cases} \qquad t_{1,j} = \begin{cases} t_{1,1} & \text{if } j = 1, \\ 0 & \text{otherwise.} \end{cases}$$

This type of matrix comes from the application in signal restoration [38, 47]. The matrices E and $\bar{T}$ are constructed using the MATLAB function `toeplitz()`. We take $\omega = 5$ and $\alpha = 1.25$. The right-hand side b is also obtained by the built-in MATLAB functions with $b = \texttt{ones}(100, 1) + \texttt{randn}(100, 1)$. All of the three methods work, and our RQI-SPCGTLS converges in comparable runtime with the RQI-PCGTLS-MP, namely, faster than the original one. Then, we turn to a bigger and more overdetermined 10000×500 random example with $\kappa(A) = 10^{10}$. We generate A with logarithmically spaced singular values and Haar random singular vectors. Sadly, neither the RQI-PCGTLS nor the RQI-PCGTLS-MP (with single PCG working precision, half Cholesky decomposition precision, and double RQI working precision) converges, while our RQI-SPCGTLS still works well.

As a second timing example, we test the RQI-SPCGTLS and the RQI-PCGTLS-MP on problems from the SuiteSparse Matrix Collection [11]. We consider the rectangular problems in the `JGD_BIBD` group for combinatorial settings, transposing wide

TABLE 1
Timings for the RQI-SPCGTLS and the RQI-PCGTLS-MP.

Matrix	**m**	**n**	**nnz**	**SPCG**	**PCG-MP**	**Speedup**
bibd_12_5	792	66	7920	0.02	0.03	1.5
bibd_17_8	24310	136	680680	0.04	0.09	2.3
bibd_20_10	184756	190	8314020	5.94	9.88	1.6
bibd_22_8	319770	231	8953560	6.38	20.72	3.3

matrices so that they are all tall. All of the matrices considered are numerically full-rank. The timing results are shown in Table 1. As these results show, the RQI-SPCGTLS can be a more efficient solver up to 3.3× speedup compared to the RQI-PCGTLS-MP.

Note that the performance of the RQI itself is even more complicated when encountering different sparsity patterns, let alone when used together with the randomized preconditioning. For example, all three methods performed poorly on the matrices in group `Mittelmann` of the Matrix Collection, while when put in the least squares settings, the randomized least squares solvers are accurate and faster than the classic ones [19]. This gap again demonstrates the difficulty of the TLS problem compared to the LS problem, and better solutions will be explored future research.

For the third experiments, we use the real-world dataset SUSY from the UCI Machine Learning Repository [2]. We consider the restricted kernel ridge regression (KRR), a supervised learning technique that uses a kernel to model nonlinearity in the data [42]. Briefly, suppose we have m data points and sample n of them, denoting the sample set as S. We then resort to the loss function of

$$\text{Loss}(x) = \|b - Ax\|^2 + \mu x^\top H x, \text{ where } a_{ij} = K(z_i, z_{s_j}), h_{ij} = K(z_{s_i}, z_{s_j}).$$

We take $\mu = 0$ here for similarity and adopt the same setup as in [18], namely, we choose the kernel function as

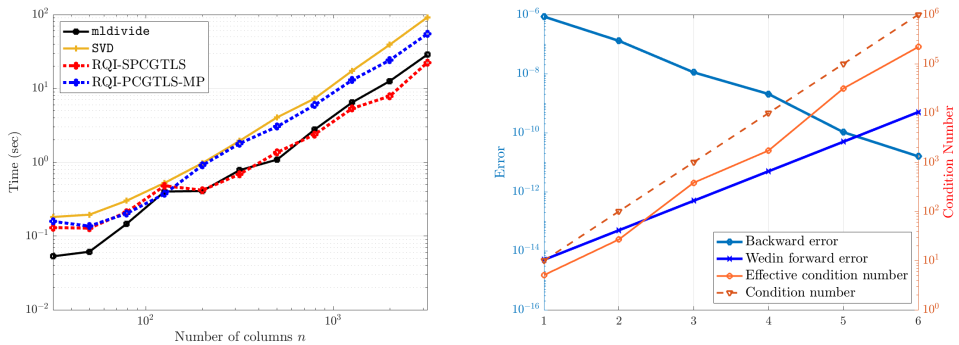
$$K(z, z') = \exp\left(-\frac{\|z - z'\|^2}{2\sigma^2}\right), \text{ with } \sigma = 4.$$

The difference is that we attempt to regard the problem in the total least squares sense. We fix the row number $m = 10^5$ and varied the column number $n \in [10^1, 10^3]$. Take the first column of the SUSY dataset as the right-hand side b and run the following four methods: `mldivide` in MATLAB, i.e., still solving the LS; the direct method SVD for the TLS; the RQI-PCGTLS-MP; and our RQI-SPCGTLS.

As shown in Figure 3, our method demonstrates increasing advantages in large-scale problems, and the complexity of $\mathcal{O}(mn\log(n/u) + n^3)$ is verified. We hope this will inspire more interdisciplinary research, since the TLS may show great merit in applications to fitting the real-world data. Finally, to illustrate the conjecture we mentioned in Remark 2.10, we create nearly compatible demo systems with changing condition numbers and plot the backward error by our TLS solver, with the Wedin forward error [52], i.e.,

$$\|x - \hat{x}\| \le 2.23\kappa(A)\left(\|x\| + \frac{\kappa(A)}{\|A\|}\|b - Ax\|\right)u,$$

of the LS solutions as a reference line. The tail of Figure 3b shows that the TLS solver can reach a level of quality almost as good as that of a forward stable LS solver.



(a) SUSY dataset (b) Fairly compatible random systems

FIG. 3. (a) *is the runtime comparison for kernel ridge regression problems between the SVD, the RQI-PCGTLS-MP, and the RQI-SPCGTLS.* (b) *illustrates the backward error and the Wedin forward error by* 500×20 *systems with moderate residuals and changing condition numbers.*

6. Conclusion. This paper establishes the theoretical analysis for the proximity of a backward stable LS solution to the true solution of its TLS problem. We identify several key conditions for efficiently solving a TLS problem and reveal their strong connections to the true LS residual and the effective condition number, highlighting the informativeness embedded in the LS structure. Moreover, a newly derived optimal backward error expression is formulated and illustrated. *Overall, we aim to shed light on the intrinsic relationship between LS and TLS problems and to offer a new perspective for leveraging well-established LS solutions in the development of efficient TLS solvers.* As an application, we propose a fast randomized iterative algorithm. Sketching preconditioners are proven to be effective, and our tested RQI-SPCGTLS is shown to be efficient. We expect this work to inspire more interdisciplinary research. For future work, we will generalize and refine the underlying theory, including the optimal backward error and its computable formulation, for multidimensional TLS problems. Note that relevant work for LS problems has been done in [43]. Addressing issues in iterative solvers, such as compatibility constraints and converging behaviors, remains an open challenge.

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